

Sustainable Software Engineering - Fairness



Ethical Artificial Intelligence: What It Is About

Machine learning systems need to live in a real environment and serve consumers. As such, their application needs to meet ethical regulations to be actually used.

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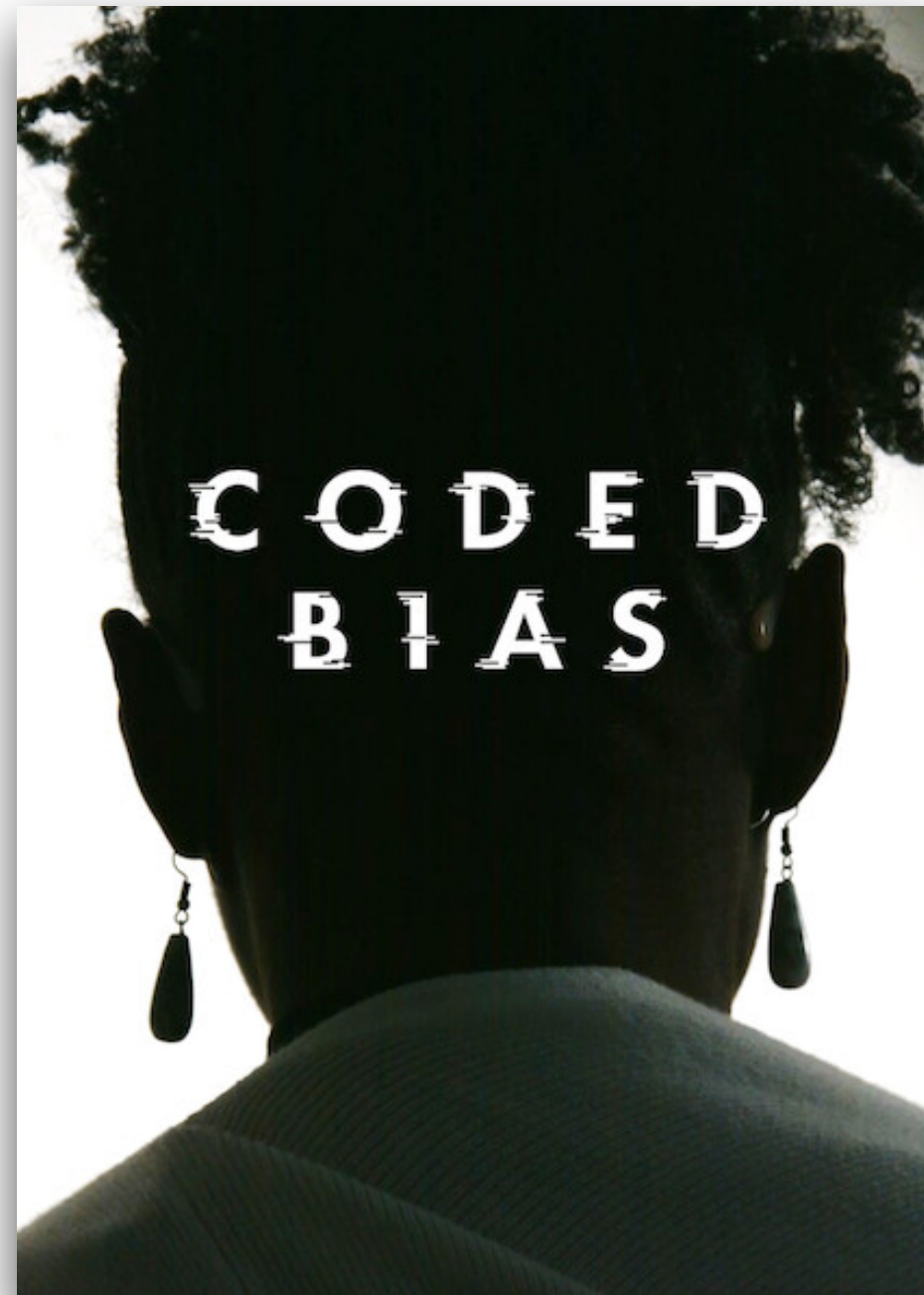
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To some extent, ethics is more important than legal matters: high ethical standards may yield long-term benefits through image and loyalty.

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The real-world relevance of ethical machine learning is proved by the amount of movies and tv series on the matter.



and many others...

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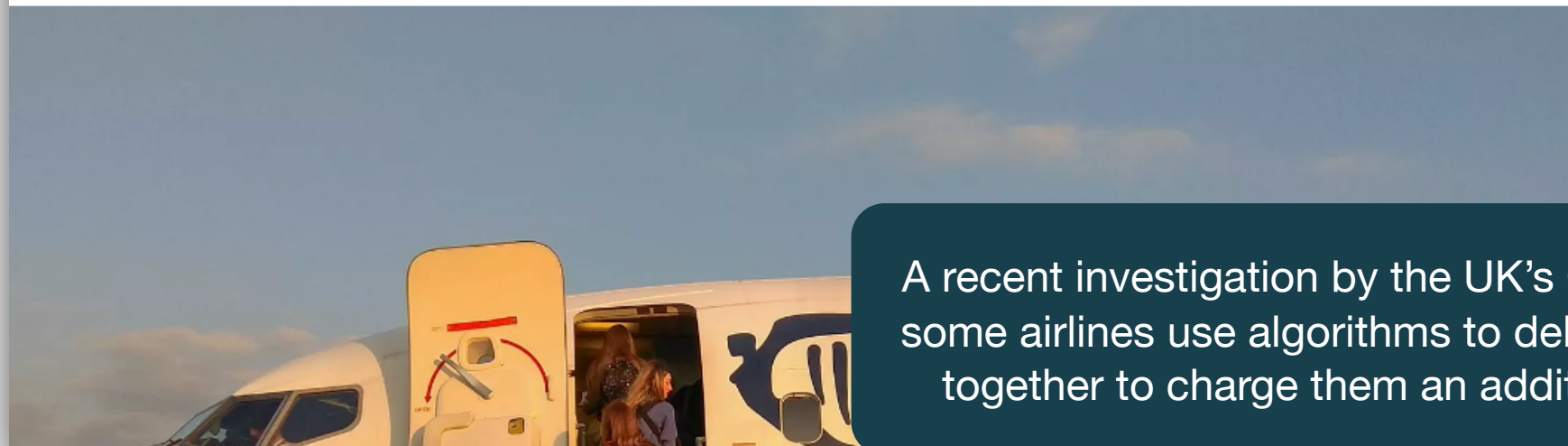
Some airlines may be using algorithms to split up families during flights

Your random airplane seat assignment might not be random at all.

By Aditi Shrikant | aditi@vox.com | Nov 27, 2018, 6:10pm EST



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A recent investigation by the UK's Civil Aviation Authority found that some airlines use algorithms to deliberately split up people traveling together to charge them an additional fee to change their seats.

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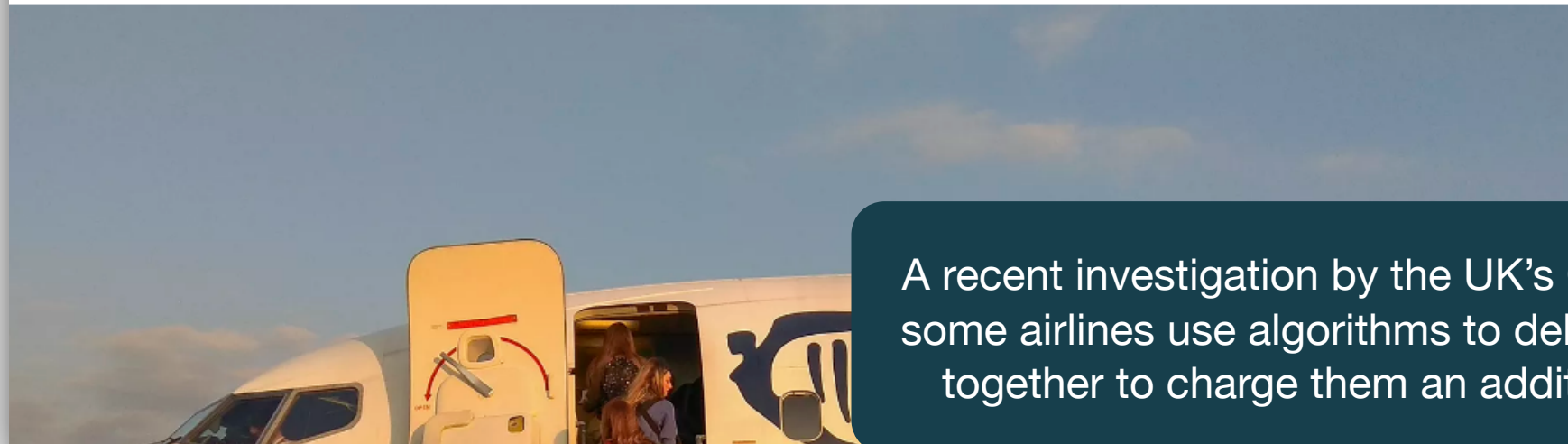
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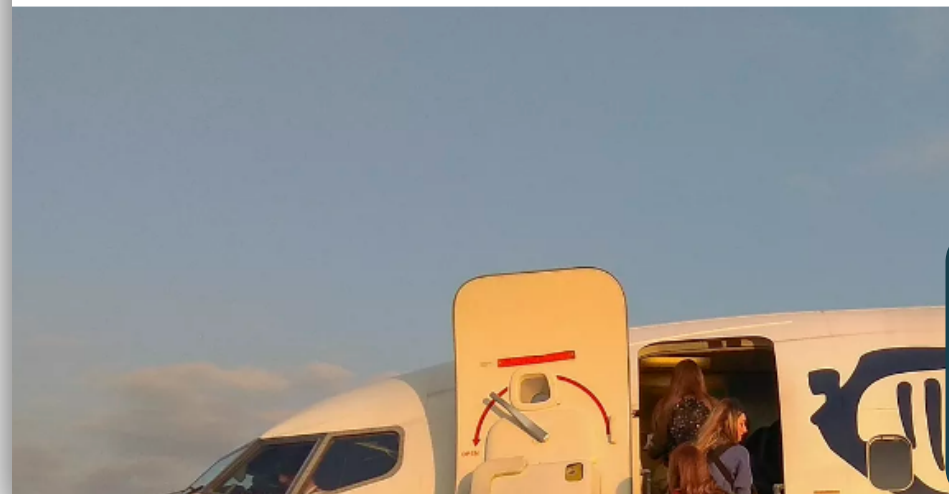
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This is just to say that the boundary between what is legal and what is ethical is often blurred and hard to draw.

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Dr. Emily Slackerman Ackerman

@EmilyEAckerman

i (in a wheelchair) was just trapped *on* forbes ave by one of these robots, only days after their independent roll out. i can tell that as long as they continue to operate, they are going to be a major accessibility and safety issue. [thread]

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Everything we know about the Starship food delivery robot
The white, 2-foot tall battery-powered delivery robots will be rolling on the sidewalk with Oakland pedestrians starting sometime in

Machine learning systems do not only revolve around algorithms, but entire internet-of-things devices. What does this imply for society and how are we dealing with it?

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Even in this case, the boundary is quite blurred. A IoT device is not supposed to threaten accessibility/safety, but in practice it may happen.

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NO MERCY NO MALICE

Robinhood Has Gamified Online Trading Into an Addiction

Tech's obsession with addiction will hurt us all



Scott Galloway [Follow](#)

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Warning: This post contains a discussion of suicide.

Addiction is the inability to stop consuming a chemical or pursuing an activity although it's causing harm.

I engage with almost every substance or behavior associated with addiction: alcohol, drugs, coffee, porn,

But machine learning may impact even differently on society, especially when used in trading activities.

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Gamification is a recent trend that aims at embedding machine learning algorithms within games so that people get engaged more.

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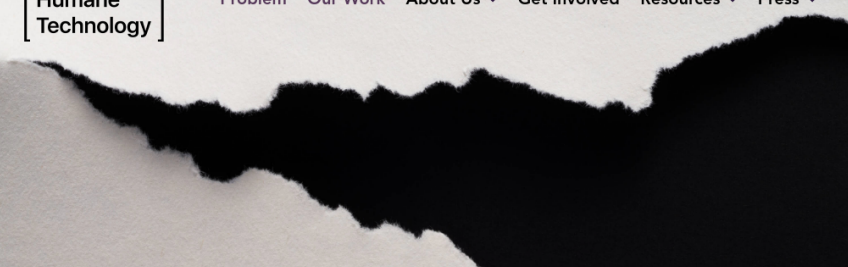
The FOMO Is Real: How Social Media Increases Depression and Loneliness

Written by [Gigen Mammoser](#) on December 10, 2018

New research reveals how social media platforms like

Center for Humane Technology

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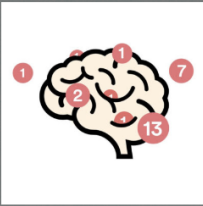
The extractive attention economy is tearing apart our shared social fabric.

The companies that created social media and mobile tech have benefited our lives enormously. But even with the best intentions, they are under intense pressure to compete for attention, creating invisible harms for society:



Digital Addiction

Digital slot machines occupy more and more space in our lives



Mental Health

We constantly face our attention, social media, and bullying



And even more... Mental health

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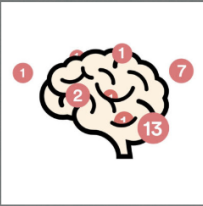
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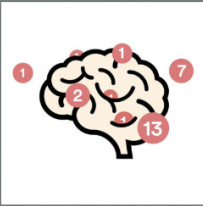
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And even more... Mental health



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Facebook Executives Shut Down Efforts to Make the Site Less Divisive

The social-media giant internally studied how it polarizes users, then largely shelved the findings

By [Jeff Horwitz](#) and [Deepa Seetharaman](#)
May 26, 2020 11:38 am ET

Polarization

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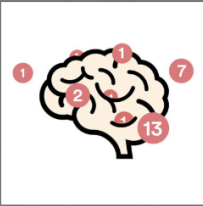
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The Washington Post

Democracy Dies in Darkness

PostEverything • Perspective

How U.S. surveillance technology is propping up authoritarian regimes

(iStock)

By [Robert Morgus](#) and [Justin Sherman](#)

Jan. 17, 2019 at 6:00 a.m. EST



Surveillance

TECH

Facebook Executives Shut Down Efforts to Make the Site Less Divisive

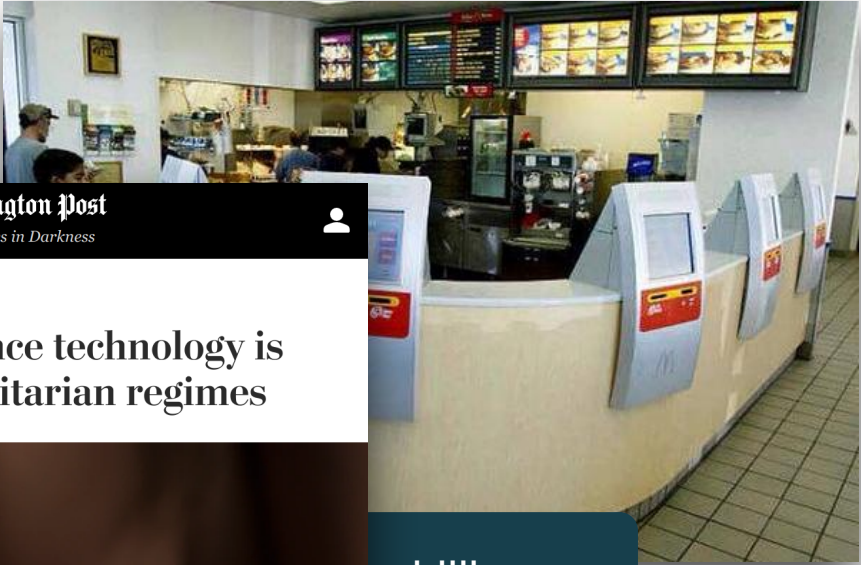
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While we cannot know what happened, we should be aware of the existence of legally protected classes.

- Color (Civil Rights Act of 1964);
- Race (Civil Rights Act of 1964);
- Sex (Equal Pay Act of 1963; Civil Rights Act of 1964);
- Religion (Civil Rights Act of 1964);
- National origin (Civil Rights Act of 1964);
- Citizenship (Immigration Reform and Control Act);
- Age (Age Discrimination in Employment Act of 1967);
- Pregnancy (Pregnancy Discrimination Act);
- Familial status (Civil Rights Act of 1968);
- Disability status (Rehabilitation Act of 1973; Americans with Disabilities Act of 1990);
- Veteran status (Vietnam Era Veterans' Readjustment Assistance Act of 1974);
- Genetic information (Genetic Information Nondiscrimination Act).

FAIRNESS AND MACHINE LEARNING

Limitations and Opportunities

Solon Barocas, Moritz Hardt, Arvind Narayanan

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- Employment (Civil Rights Act of 1964);
- Housing (Fair Housing Act);
- Public Accommodation (Civil Rights Act of 1964).

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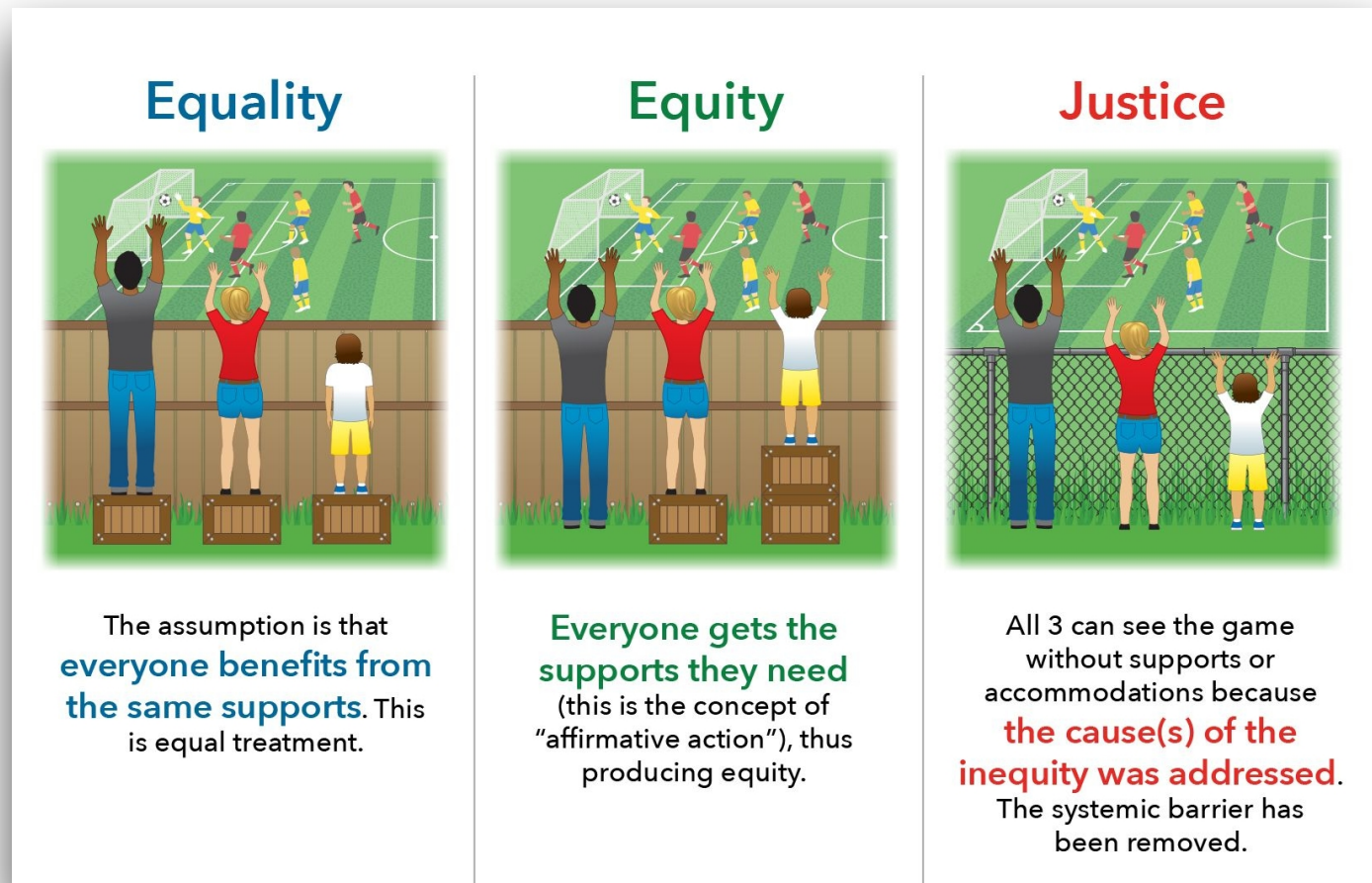
But also of the various levels of actions you may take.

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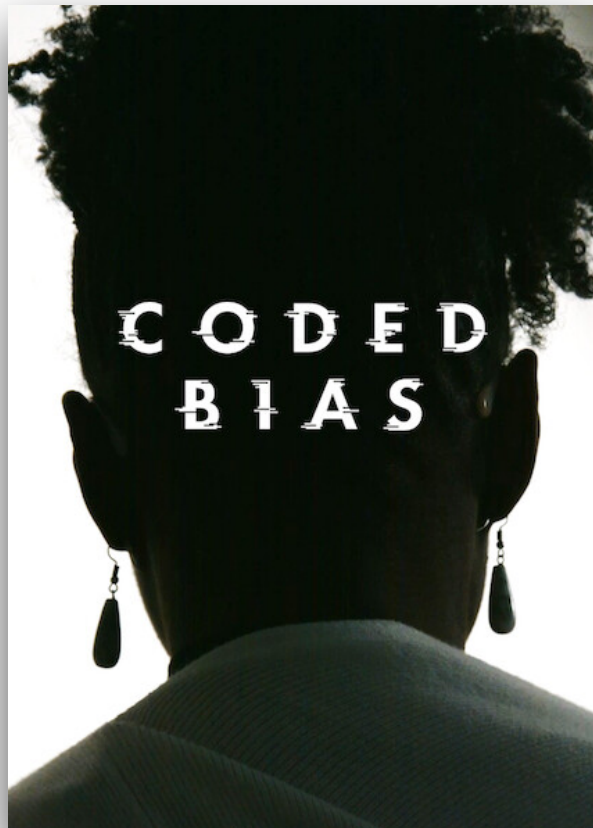
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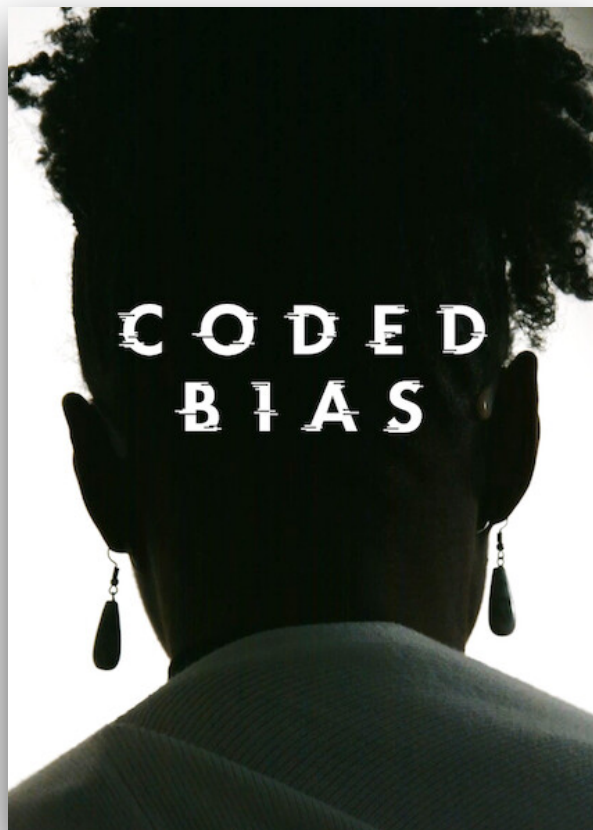
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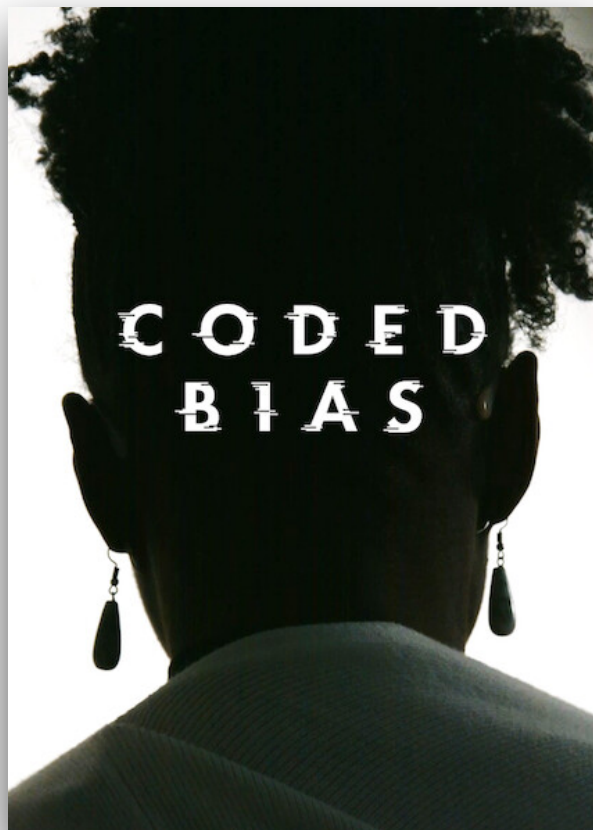
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SOFTWARE MAINTENANCE AND EVOLUTION INTERMEZZO - DID YOU KNOW?

Computer Science has been often considered as a typical man activity: this is a huge historical forgery. *Margaret Hamilton* was likely the first software engineer. You might have heard about her already: she was the leader of the team involved in the NASA Apollo program that led to the landing of the first humans on the Moon.

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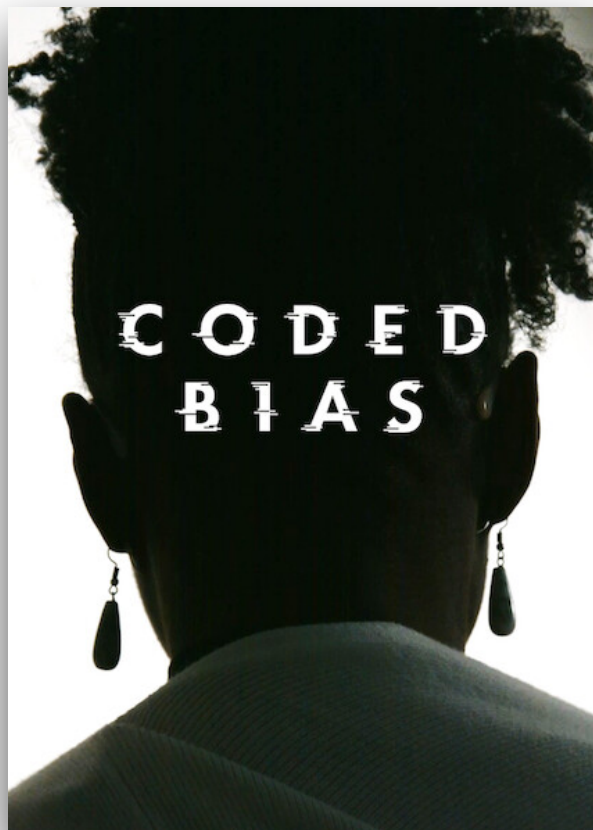
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Long story short: Voluntarily or not, data can perpetuate society’s existing race-, class- and gender-based inequities.

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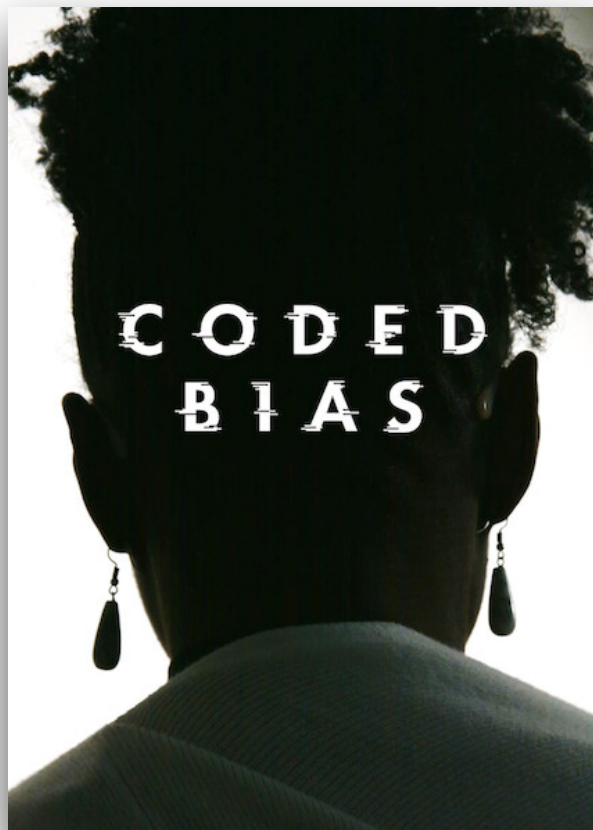
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“When you think of A.I., it’s forward-looking. But A.I. is based on data, and data is a reflection of our history.”

Artificial intelligence applications use previous data to make predictions or find patterns that will be used in the future, which may be sometimes problematic.

In this sense, ethical concerns are embedded in the project solution domain. Some issues arise by design (others may be even maliciously coded). Computer scientists: if (longly) historically associated to men, how should an AI work?

Long story short: Voluntarily or not, data can perpetuate society’s existing race-, class- and gender-based inequities.

Capita Selecta of Software Engineering: Ethics and Fairness

Ethical Artificial Intelligence: What It Is About

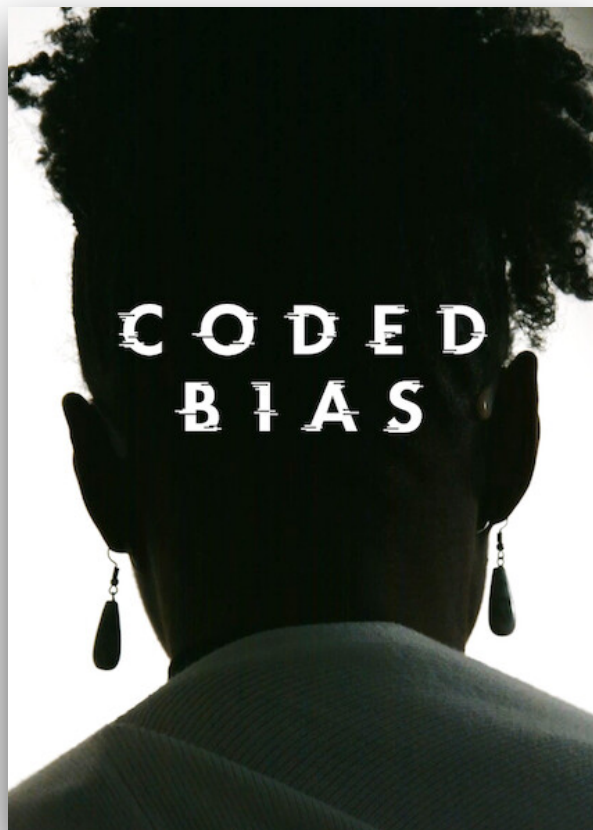
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Unequal treatment, unequal outcomes obtained by reinforcing patterns in predicting policing with feedback loops.

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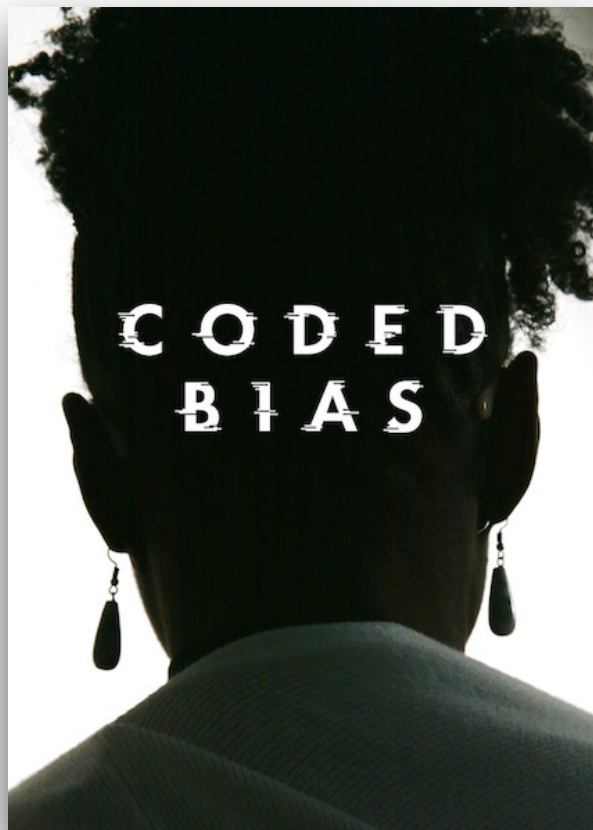
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And always remember: Everyone is biased toward something! Our culture, our education, and others, influence us in many, many hidden things.

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The former refers to algorithmic issues, the latter to the distinguishing outcomes.

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Also consider that predictions are made on correlations, not causation! They may depend on flawed statistics or deteriorated by feedback loops.

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Given the contextual nature of software fairness, more than twenty different definitions have been provided so far, based on statistical, similarity, and casual reasoning terms.

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- (5) We refer to the *prediction* as the decision taken by an algorithm with respect to the category to assign to a certain individual.

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Based on these concepts, we can reason on the implementation of the fairness concept in pure statistical terms.

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Group fairness, a.k.a. Statistical parity

According to this definition, a binary machine learner is fair if individuals of a protected group G - defined based on a protected attribute p - are associated to the positive class with the same probability of assigning the individuals of non-protected groups.

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These are just some of the definitions - many others come as we speak! In any case, it may be a good idea that of combining multiple definitions to have a better fairness idea.

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The statistical definitions suffer from a key issue: all are based on the concepts of *protected groups*, i.e., they all define fairness solely based on the protected attributes without considering other implementation constraints.

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Casual discrimination

According to this definition, a binary machine learner is fair if it produces the same classification for any two subjects with the exact same attributes X . For instance, this implies that a male and female applicants who otherwise have the same attributes X will either both be hired or not.

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The similarity-based definitions are therefore more holistic than the statistical ones.

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Causal reasoning definitions assume a given causal graph: a directed, acyclic graph with nodes representing attributes of an applicant and edges representing relationships between the attributes.

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In practice, defining these relations is hard or even impossible. This at a deep learner configured with multiple hidden layers: how can you describe cause-effect relations?

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Anyway, provided a causal graph of this type, you may define fairness in more precise manners, such as:

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So, depending on the definition of fairness, we might end up considering an artificial intelligence solution as fair or unfair.

Ethical Artificial Intelligence: What It Is About (in Practice)

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But this is not just a matter of definitions. Most of the established metrics assume availability of actual, verified outcomes. These outcomes are available for the training data, yet it is unclear if the real classified data always conforms to the same distribution.

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Abstract: This dataset classifies people described by a set of attributes as good or bad credit risks. Comes in two formats (one all numeric). Also comes with a cost matrix

Data Set Characteristics:	Multivariate	Number of Instances:	1000	Area:	Financial
Attribute Characteristics:	Categorical, Integer	Number of Attributes:	20	Date Donated	1994-11-17
Associated Tasks:	Classification	Missing Values?	N/A	Number of Web Hits:	795669

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Capita Selecta of Software Engineering: Ethics and Fairness

Ethical Artificial Intelligence: What It Is About (in Practice)

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Based on the definition of fairness you choose, you might end up considering a classifier to act fairly or not. Read more here: <https://fairware.cs.umass.edu/papers/Verma.pdf>.

Ethical Artificial Intelligence: Dealing With Fairness

And so, how can we deal with software fairness using software engineering methods?

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Anti-classification or Fairness through Blindness

Ignore/eliminate sensitive attributes from dataset, e.g., remove gender and race from a credit card scoring system.

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Classification parity

Classification error is equal across groups.

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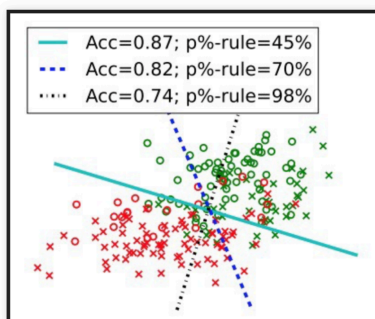
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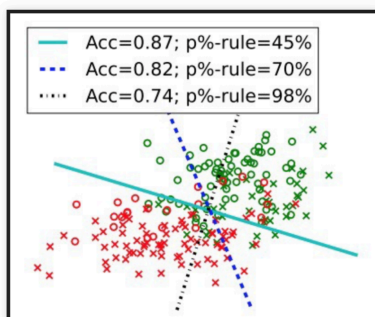
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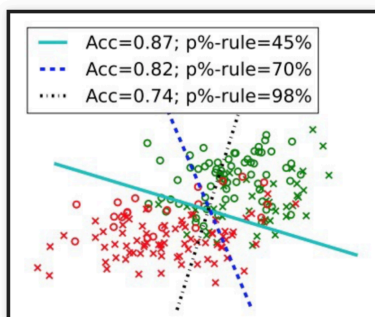
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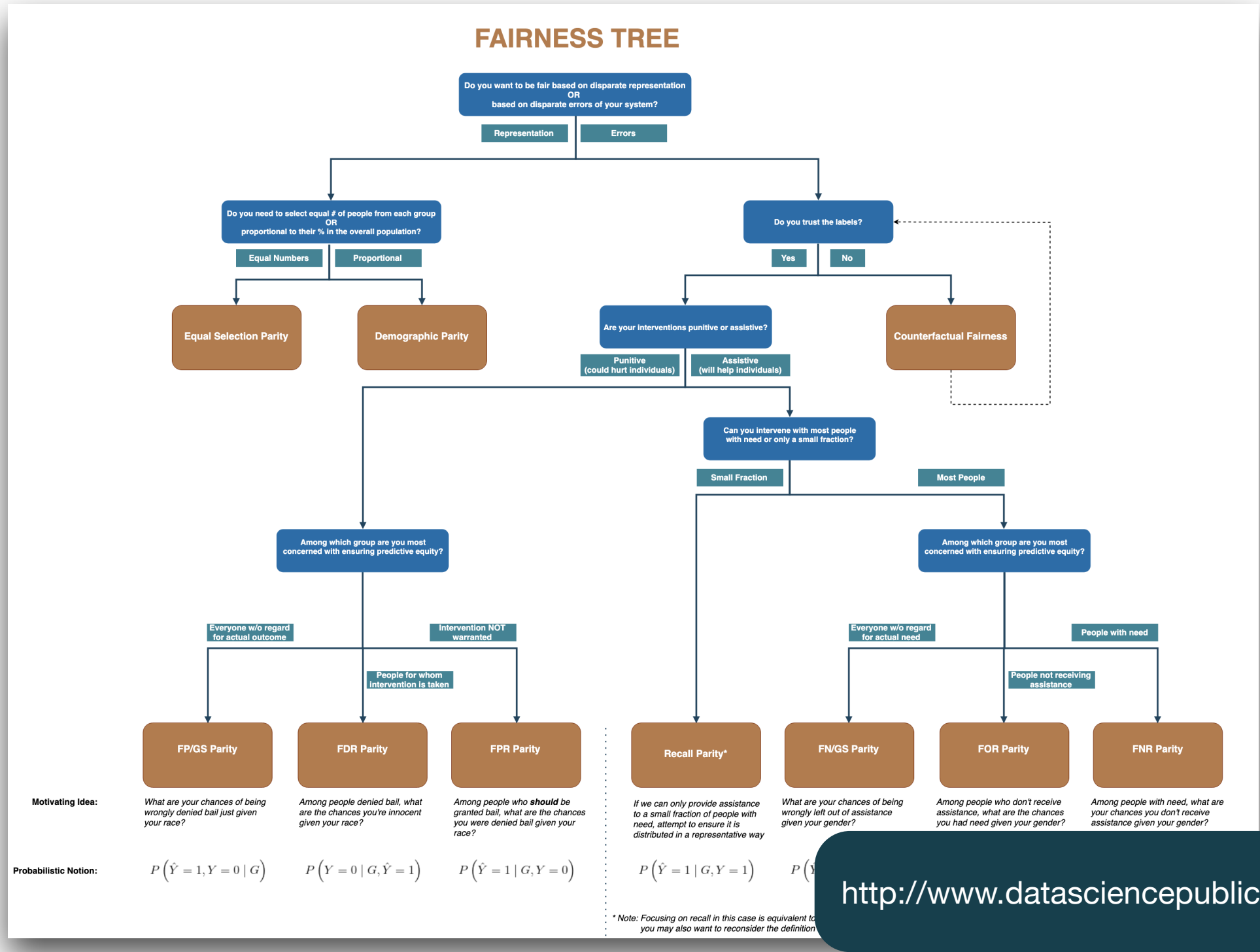
Post-processing

Adjust the learned model to be uncorrelated with sensitive attributes, for instance by testing multiple ways to set hyper-parameters.

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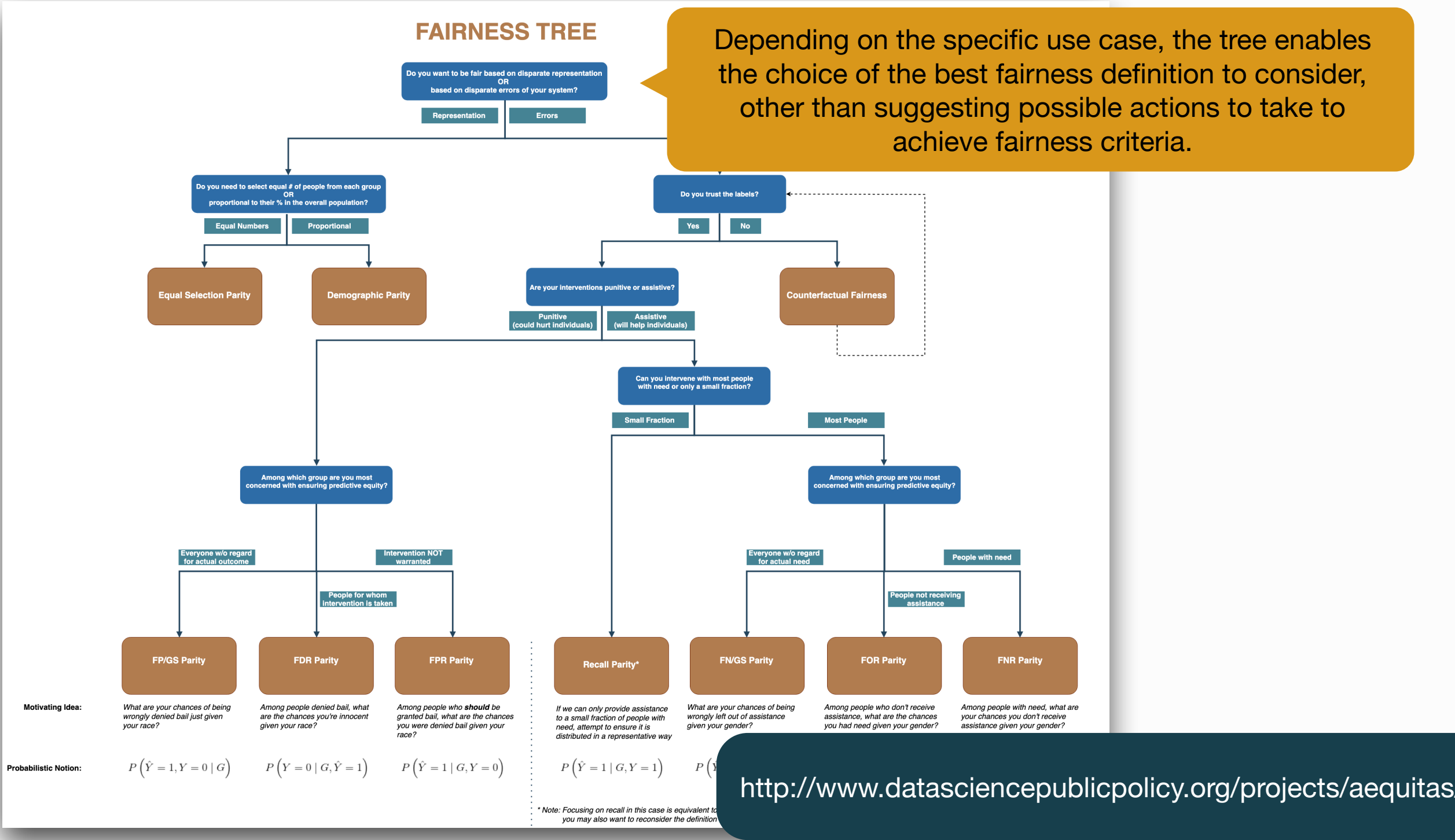
Assessing fairness on-the-fly is possible. Aequitas is an open-source audit toolkit for data scientists and software engineers. It implements the so-called fairness tree:



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- **Functional:** biases due to platform affordances and algorithms
- **Normative:** biases due to community norms
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- **Acquisition:** biases due to, e.g., API limits
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- **Cleaning:** biases due to, e.g., default values
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Data Analysis



- **Qualitative Analyses:** lack generalizability, interpret. biases
- **Descriptive Statistics:** confounding bias, obfuscated measurements
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Evaluation

- **Metrics:** e.g., reliability, lack of domain insights
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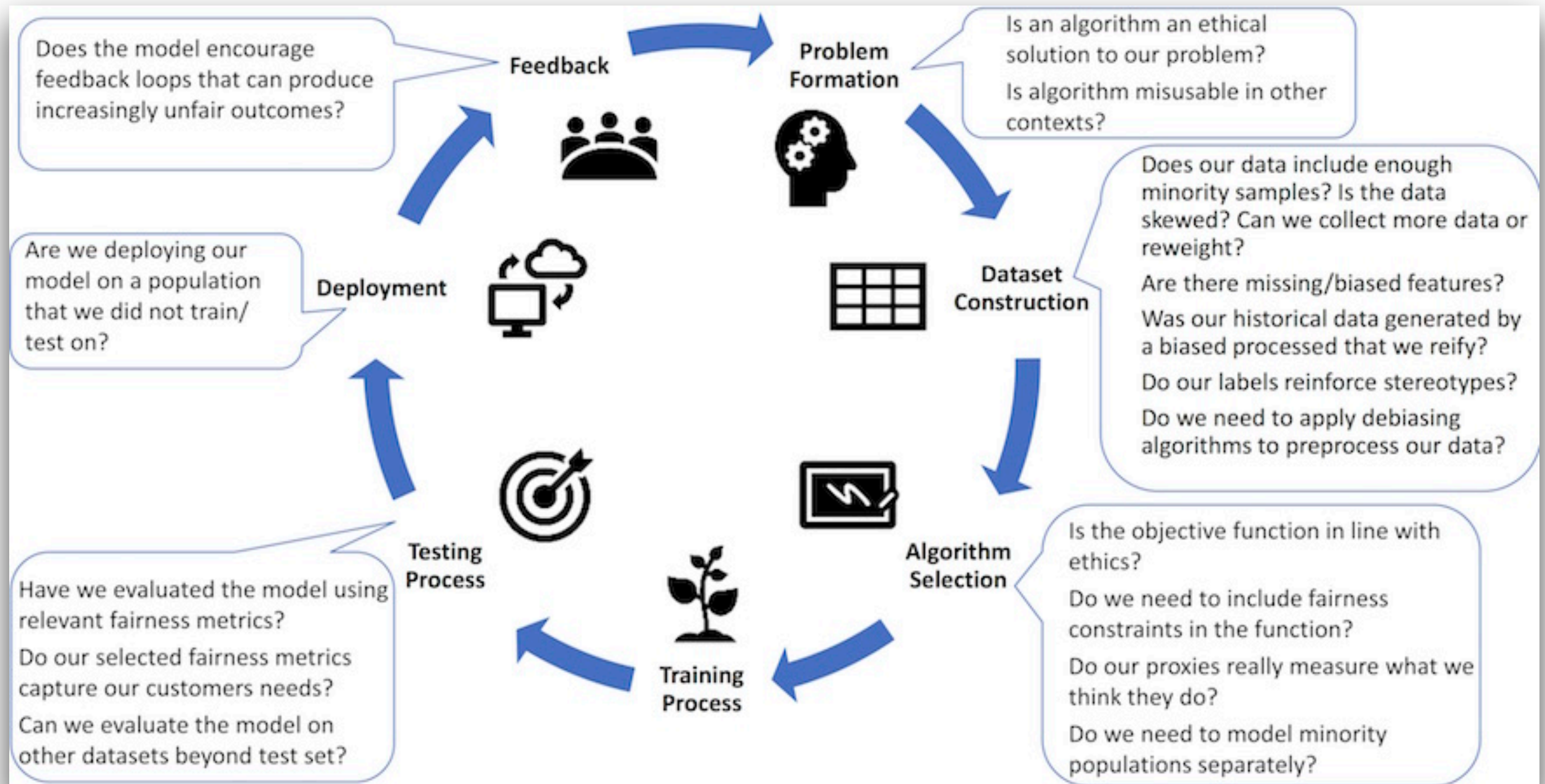
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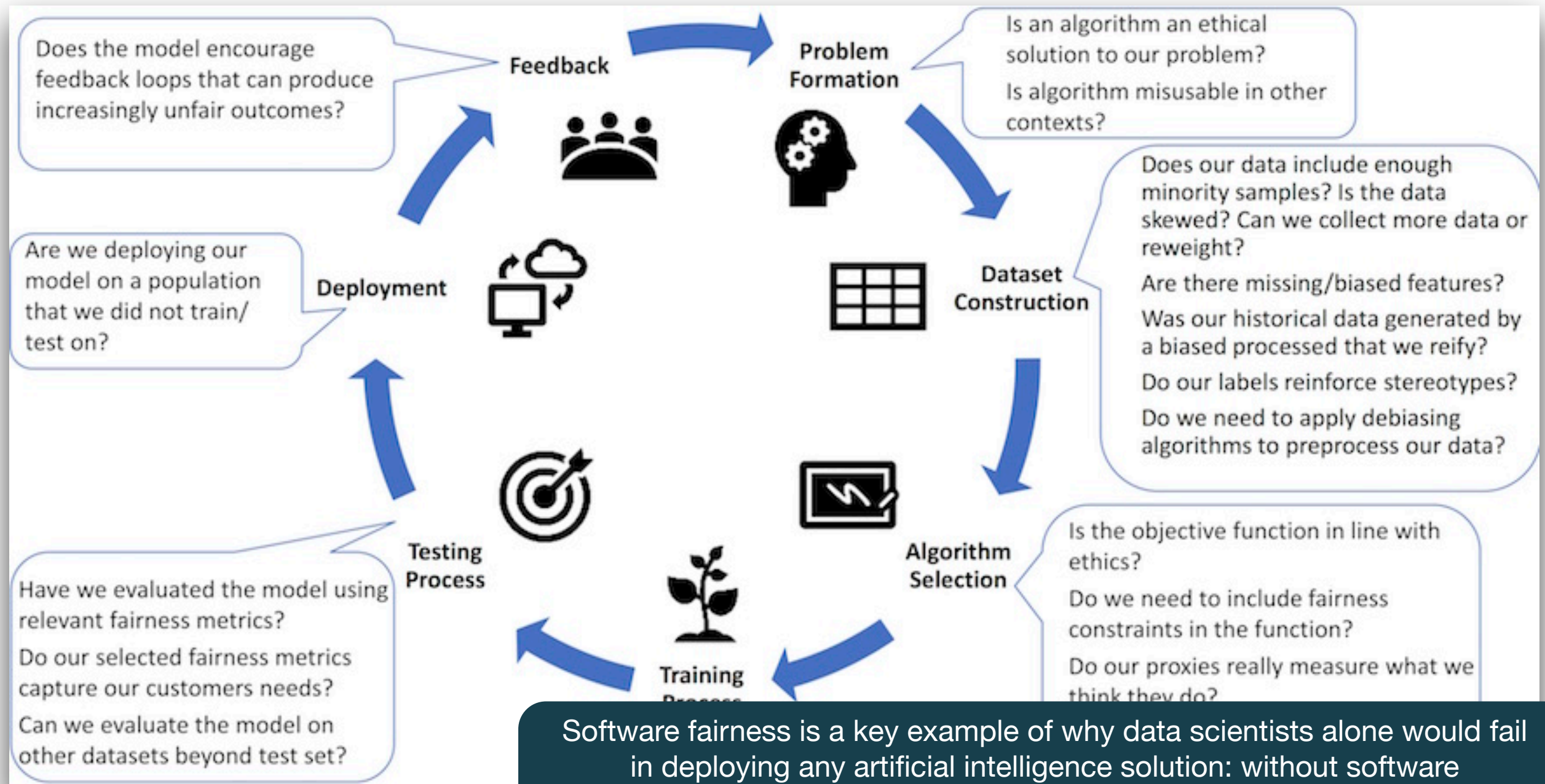
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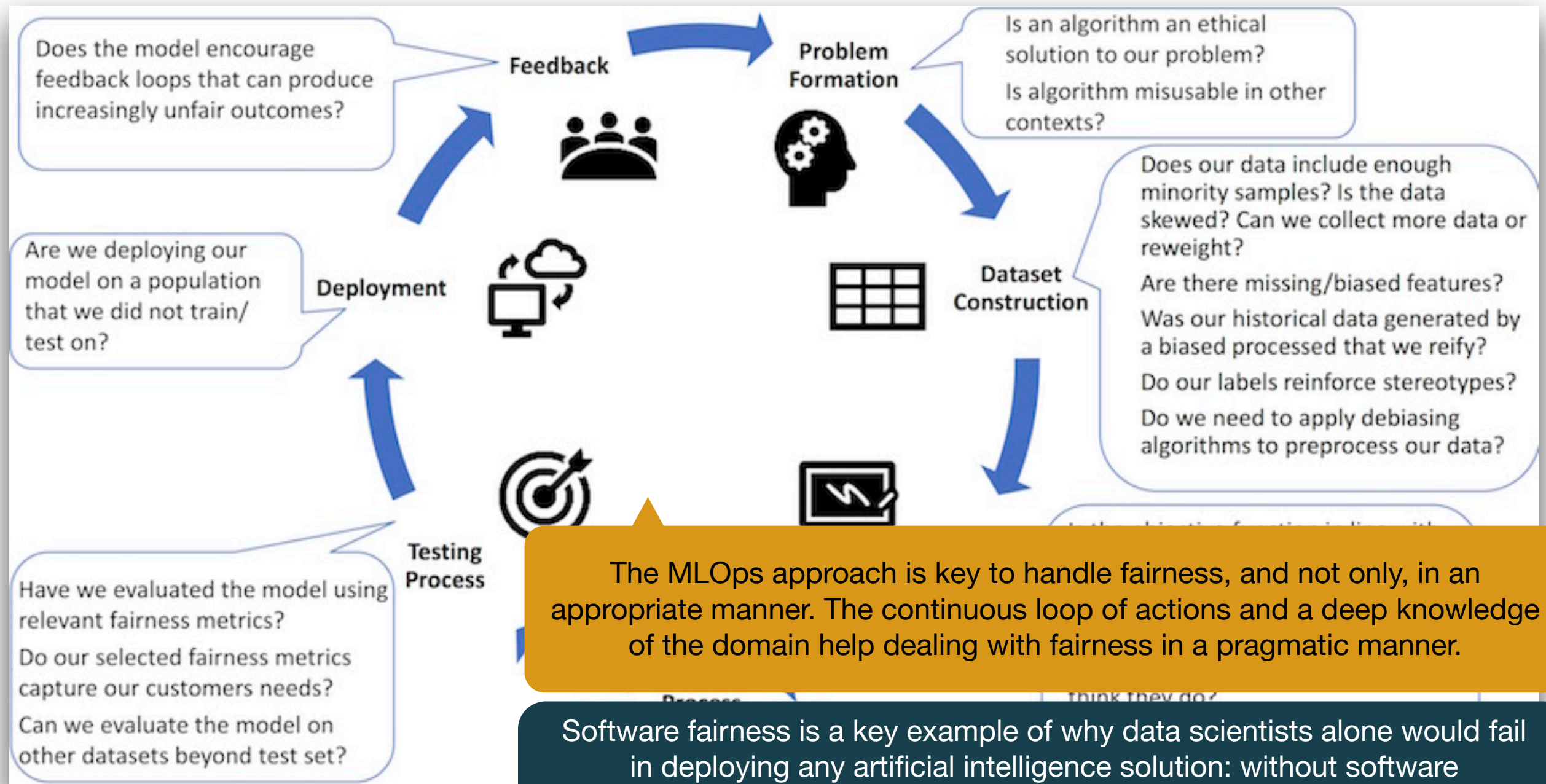


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The MLOps approach is key to handle fairness, and not only, in an appropriate manner. The continuous loop of actions and a deep knowledge of the domain help dealing with fairness in a pragmatic manner.

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Sustainable Software Engineering - Fairness

